

Luca Mancuso

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TECHNICAL SKILLS

- **Programming Languages:** JavaScript, Java, Python, SQL, C, C++, C#, TypeScript, HTML/CSS, R, PHP, GDScript
- **Libraries and Frameworks:** React, Next.js, Tailwind CSS, Node.js, Express, Prisma, Vite
- **Database Systems:** MySQL, MongoDB, Firebase Firestore, Supabase
- **Developer Tools:** Linux, Git, VS Code, Jupyter Notebook, MongoDB Compass, AI Pair Programming (Claude)
- **Core Competencies:** Data Structures, Debugging, Software Development, System Design and Architecture

PROJECTS

Full Stack Web App – MuzicBoxd

muzicboxd.netlify.app

- Architected and directed full-stack development of a letterboxd-inspired music journaling web app, using AI pair-programming (Claude) to accelerate implementation while personally owning system design, data modeling, and API integration decisions
- Designed the data layer around the iTunes Search API's collectionId as the album identifier after rejecting MusicBrainz for reliability issues
- Built and deployed a production stack: React/Vite/TypeScript frontend (Netlify), Node.js/Express backend (Render), PostgreSQL via Neon with Prisma ORM, and Google OAuth via Passport.js
- Implemented a dated journalling system and profile analytics so users can keep track of their listening habits

Godot Game – Spell It Out!

[GitHub demo available](#)

- Designed and solo-developed a Scrabble & Balatro-inspired roguelike word game in Godot/GDScript over a five-month development cycle
- Built complex systems from scratch including scoring and combo logic, event trigger orders, and an upgrade economy with a rarity weighted shop, then iterated on game balance and UX using player feedback across 5 beta builds
- Instrumented logging to debug system issues, and fixed critical bugs to ensure a stable and reliable player experience

Full Step Web App – Gridz

albumgridz.netlify.app

- Designed and built a full-stack drag-and-drop grid builder for creating, naming, and saving custom album collages ("topsters"), distinct from Muzicboxd in both interaction model and data shape
- Implemented drag-and-drop grid placement using React DnD, with album search and metadata via the Last.fm API
- Used Firebase Authentication for secure login and Firestore to persist and retrieve each user's saved topsters under their account

EDUCATION

University of Western Ontario

Faculty of Science, BSc in Computer Science

Expected graduation 2026

London, ON

ACCOMPLISHMENTS

LoJam x Western Game Development Society Hackathon/GameJam 2025 – Platformer Game

- Competed as a solo entrant, achieving 8th place out of a large pool of contestants
- Challenged with creating and presenting a game in one week, learned the Godot game engine and its language GDScript to create a game that satisfied the competition's theme in a creative and meaningful way
- Created the game's concept, node scripts, physics logic, game art, sound effects, and user interface
- Presented and gave a live demonstration of my work to all the entrants and judges

AIInnovation Partners 2023 ThinkATHon Competition – Artificial Intelligence Based Proposal

- Collaborated with a team to achieve 2nd place out of 30 groups
- Proposed an AI-based business venture to secure investment from venture capitalists
- Researched the competitive landscape and a viable go-to-market strategy using AI in telemarketing